S8 Table. Subgroup analyses of case fatality rate and prevalence of yellow fever virus in humans in Africa.

	Prevalence. % (95%CI)	95% Prediction interval	N Studies	N Participants	H (95%CI)	I^2 (95%CI)	P heterogeneity	P difference subtypes
YFV case fatality rate in humans								
Current infection								
Study Design								0.35
Cross sectional	23.1 [14.3-33.1]	NA	1	78	NA	NA	1	
Outbreak	37.2 [8.5-71.1]	NA	2	23	1.6 [1-3.3]	60.2 [0-90.7]	0.113	
Setting								0.032
Community-based	53.8 [26.1-80.5]	NA	1	13	NA	NA	1	
Hospital-based	23.1 [14.3-33.1]	NA	1	78	NA	NA	1	
Countries								0.094
Democratic Republic of the Congo	23.1 [14.3-33.1]	NA	1	78	NA	NA	1	
Nigeria	20 [0.5-51.3]	NA	1	10	NA	NA	1	
Uganda	53.8 [26.1-80.5]	NA	1	13	NA	NA	1	
UNSD Region								0.094
Central Africa	23.1 [14.3-33.1]	NA	1	78	NA	NA	1	
Eastern Africa	53.8 [26.1-80.5]	NA	1	13	NA	NA	1	
West Africa	20 [0.5-51.3]	NA	1	10	NA	NA	1	
Country income level								0.512
Low-income economies	34.9 [9-66.4]	NA	2	91	2.1 [1-4.5]	78.3 [5.5-95]	0.032	
Lower-middle income economies	20 [0.5-51.3]	NA	1	10	NA	NA	1	
YFV prevalence in humans								
Current infection								
Study Design								0.055
Cross sectional	3.6 [1.3-6.8]	[0-20.8]	12	21313	7.7 [6.8-8.8]	98.3 [97.8-98.7]	<0.001	
Outbreak	10.6 [3.6-20.2]	[0-50.5]	7	940	3.6 [2.7-4.7]	92.1 [86.2-95.4]	<0.001	
Sampling								<0.001
Non probabilistic	6 [3.2-9.6]	[0-25.9]	18	21817	6.8 [6.1-7.6]	97.9 [97.3-98.3]	<0.001	
Probabilistic	0 [0-0.4]	NA	1	436	NA	NA	1	
Setting								0.196
Community-based	8 [2.4-16]	[0-42.3]	6	4480	3.9 [2.9-5.1]	93.3 [88-96.2]	<0.001	
Hospital-based	3.8 [1-8]	[0-27.4]	12	17742	7.5 [6.6-8.5]	98.2 [97.7-98.6]	<0.001	
Hospitalization								0.039

	Prevalence. % (95%CI)	95% Prediction interval	N Studies	N Participants	H (95%CI)	I^2 (95%CI)	P heterogeneity	P difference subtypes
Ambulatory	0.7 [0-4.7]	[0-100]	3	2329	3.6 [2.3-5.8]	92.5 [81.3-97]	<0.001	
Hospitalized	16.7 [1.4-42.4]	[0-100]	5	831	7.7 [6.2-9.5]	98.3 [97.4-98.9]	<0.001	
Timing of samples collection								0.276
Prospetively	6.3 [2.7-11.2]	[0-33.9]	16	17826	7 [6.2-7.8]	97.9 [97.4-98.4]	<0.001	
Retrospectively	2.7 [0-8.7]	[0-100]	3	4427	4.4 [2.9-6.6]	94.8 [88.2-97.7]	<0.001	
Countries								<0.001
Burkina Faso	9.5 [0-47.9]	NA	2	3991	5.2 [3.1-8.7]	96.3 [89.8-98.7]	<0.001	
Côte d'Ivoire	0.7 [0.1-1.7]	NA	2	448	1 NA	0 NA	0.883	
Democratic Republic of the Congo	2.9 [0.3-7.6]	[0-36.9]	4	2672	3.8 [2.6-5.5]	92.9 [85.1-96.6]	<0.001	
Ethiopia	0 [0-0.9]	NA	1	200	NA	NA	1	
Gabon	0 [0-0.4]	NA	1	436	NA	NA	1	
Nigeria	45.6 [21.8-70.6]	NA	2	81	2.2 [1.1-4.6]	80.1 [14.4-95.4]	0.025	
Senegal	0 [0-0]	NA	2	13949	1 NA	0 NA	0.47	
South Sudan	17.9 [5.5-34.5]	NA	1	28	NA	NA	1	
Sudan	11.1 [0-64.3]	NA	2	253	9.5 [6.7-13.5]	98.9 [97.7-99.5]	<0.001	
Uganda	12.5 [1.3-31]	NA	2	195	2 [1-4.3]	75.9 [0-94.5]	0.042	
UNSD Region								0.37
Central Africa	1.9 [0-5.8]	[0-24.7]	5	3108	4.4 [3.3-5.9]	94.8 [90.7-97.1]	<0.001	
Eastern Africa	8 [0.2-23]	[0-91.2]	4	423	3.7 [2.6-5.4]	92.9 [85-96.6]	<0.001	
Northern Africa	11.1 [0-64.3]	NA	2	253	9.5 [6.7-13.5]	98.9 [97.7-99.5]	<0.001	
West Africa	5.5 [2.2-9.9]	[0-24.7]	8	18469	6.8 [5.7-8.1]	97.8 [97-98.5]	<0.001	
Country income level								<0.001
Low-income economies	6.1 [2.8-10.4]	[0-26.7]	12	7339	5.1 [4.3-6]	96.1 [94.6-97.2]	<0.001	
Lower-middle income economies	6.6 [0.6-17.4]	[0-58.5]	6	14478	6.3 [5.1-7.8]	97.5 [96.2-98.4]	<0.001	
Upper-middle-income economies	0 [0-0.4]	NA	1	436	NA	NA	1	
YFV vaccine								0.172
No	28.2 [16.6-41.4]	NA	2	53	1 NA	0 NA	0.472	
Yes/No	8 [0-37.7]	NA	2	420	4.4 [2.5-7.6]	94.7 [83.9-98.3]	<0.001	
Study population: Humans								<0.001
Febrile patients	0 [0-0]	[0-0]	6	14931	1 [1-2]	0 [0-74.6]	0.87	
Mixed human categories	17.9 [5.5-34.5]	NA	1	28	NA	NA	1	
YFV positive case contact	0.5 [0-2.3]	NA	1	182	NA	NA	1	
YFV suspected cases	13.4 [7.6-20.6]	[0-44.5]	11	7112	6.1 [5.3-7.2]	97.3 [96.4-98]	<0.001	
Detection assay								0.342

	Prevalence. % (95%CI)	95% Prediction interval	N Studies	N Participants	H (95%CI)	I^2 (95%CI)	P heterogeneity	P difference subtypes
Classical RT-PCR	0.9 [0.3-1.8]	[0-10.1]	3	831	1 [1-3.1]	0 [0-89.6]	0.743	
Culture	4.3 [0-35.9]	NA	2	464	4.4 [2.5-7.7]	94.9 [84.4-98.3]	<0.001	
Real Time RT-PCR	3.4 [0-10.2]	[0-38.5]	7	14567	5.3 [4.3-6.6]	96.4 [94.5-97.7]	<0.001	
Target detected								0.756
Live virus	4.3 [0-35.9]	NA	2	464	4.4 [2.5-7.7]	94.9 [84.4-98.3]	<0.001	
Viral RNA	2.1 [0.2-5.3]	[0-19.8]	10	15398	4.6 [3.8-5.6]	95.4 [93.2-96.8]	<0.001	
Past infection								
Study Design								<0.001
Cross sectional	18 [11.2-26]	[0-62.4]	21	14973	11.3 [10.5-12.2]	99.2 [99.1-99.3]	<0.001	
Outbreak	36 [32.3-39.9]	NA	1	605	NA	NA	1	
Sampling								0.016
Non probabilistic	26.2 [14.3-40.2]	[0-84.7]	14	9655	13.4 [12.4-14.5]	99.4 [99.3-99.5]	<0.001	
Probabilistic	9.4 [3.8-17.1]	[0-45.1]	8	5923	8.7 [7.5-10.1]	98.7 [98.2-99]	<0.001	
Setting								0.953
Community-based	12.9 [6.7-20.9]	[0-53.8]	14	13352	12 [11-13.1]	99.3 [99.2-99.4]	<0.001	
Hospital-based	12 [3.1-25.1]	[0-83]	4	1222	4.9 [3.6-6.7]	95.8 [92.1-97.8]	<0.001	
Hospitalization								<0.001
Ambulatory	14.4 [0.4-41.1]	[0-100]	6	1850	12.3 [10.7-14.1]	99.3 [99.1-99.5]	<0.001	
Hospitalized	95.5 [76.8-100]	NA	2	42	1.7 [1-3.6]	66.7 [0-92.5]	0.083	
Timing of samples collection								0.523
Prospetively	20 [12.1-29.4]	[0-67.7]	18	13510	12 [11.1-13]	99.3 [99.2-99.4]	<0.001	
Retrospectively	14 [2.2-33.3]	[0-99.7]	4	2068	10 [8.2-12.2]	99 [98.5-99.3]	<0.001	
Countries								<0.001
Cameroon	26.1 [12-43.1]	[0-100]	3	353	2.6 [1.5-4.5]	85.2 [56.5-95]	0.001	
Central African Republic	6.6 [2.4-12.6]	[0-100]	3	3239	5.6 [3.9-8]	96.8 [93.5-98.4]	<0.001	
Côte d'Ivoire	79.3 [26.7-100]	[0-100]	3	647	6 [4.3-8.5]	97.2 [94.6-98.6]	<0.001	
Democratic Republic of the Congo	17.1 [14.8-19.5]	NA	1	978	NA	NA	1	
Ethiopia	15.7 [0.1-48]	[0-100]	3	2091	12.1 [9.8-15.1]	99.3 [99-99.6]	<0.001	
Gabon	59.5 [55-64]	NA	1	462	NA	NA	1	
Kenya	10.1 [3.5-19.5]	[0-53.5]	6	3684	8 [6.7-9.6]	98.4 [97.7-98.9]	<0.001	
Tanzania	0.8 [0.2-1.8]	NA	1	500	NA	NA	1	
Zambia	0.3 [0.2-0.5]	NA	1	3624	NA	NA	1	
UNSD Region								0.01
Central Africa	19.1 [8.5-32.7]	[0-73.1]	8	5032	10.4 [9.1-11.8]	99.1 [98.8-99.3]	<0.001	

	Prevalence. % (95%CI)	95% Prediction interval	N Studies	N Participants	H (95%CI)	I^2 (95%CI)	P heterogeneity	P difference subtypes
Eastern Africa	8.8 [3.3-16.5]	[0-48.3]	11	9899	11.2 [10.1-12.4]	99.2 [99-99.3]	<0.001	
West Africa	79.3 [26.7-100]	[0-100]	3	647	6 [4.3-8.5]	97.2 [94.6-98.6]	<0.001	
Country income level								<0.001
Low-income economies	11.5 [5.5-19.4]	[0-46.4]	7	6308	8.5 [7.2-10]	98.6 [98.1-99]	<0.001	
Lower-middle income economies	20.8 [11-32.5]	[0-73.8]	14	8808	11.3 [10.3-12.4]	99.2 [99.1-99.3]	<0.001	
Upper-middle-income economies	59.5 [55-64]	NA	1	462	NA	NA	1	
YFV vaccine								0.436
No	7.3 [2.1-15.4]	[0-46.9]	8	8833	11.5 [10.1-12.9]	99.2 [99-99.4]	<0.001	
Yes/No	18.1 [0.2-54.7]	[0-100]	3	1995	17.6 [14.9-20.8]	99.7 [99.5-99.8]	<0.001	
Study population: Humans								<0.001
Apparently healthy individuals	14.1 [7.8-21.8]	[0-54.9]	16	13939	11.6 [10.6-12.6]	99.3 [99.1-99.4]	<0.001	
Febrile patients	23 [0-65.4]	NA	2	97	3.4 [1.8-6.3]	91.2 [68.9-97.5]	0.001	
General population	0.8 [0.2-1.8]	NA	1	500	NA	NA	1	
Pregnant women	42 [37.3-46.8]	NA	1	419	NA	NA	1	
YFV positive case contact	36 [32.3-39.9]	NA	1	605	NA	NA	1	
YFV suspected cases	100 [90.7-100]	NA	1	18	NA	NA	1	
Detection assay								<0.001
Complement fixation test	5.6 [0-22.3]	NA	1	18	NA	NA	1	
Enzyme immunoassay	6.5 [5.4-7.7]	NA	1	1762	NA	NA	1	
Hemagglutination inhibition test	43 [32.3-54.1]	NA	1	79	NA	NA	1	
Indirect ELISA	21.6 [10.9-34.6]	[0-76.8]	12	9963	13.5 [12.4-14.7]	99.5 [99.3-99.5]	<0.001	
Indirect immunofluorescence assay	21.1 [0-63.9]	NA	2	541	8.9 [6.1-12.9]	98.7 [97.3-99.4]	<0.001	
Plaque reduction neutralization test (PRNT)	20.6 [3.3-47.4]	[0-100]	3	1738	11.6 [9.3-14.5]	99.3 [98.8-99.5]	<0.001	
Target detected								0.449
Antibodies	14.7 [4.8-28.5]	[0-71.6]	7	3040	8.9 [7.6-10.4]	98.7 [98.3-99.1]	<0.001	
IgG	21 [11.9-31.7]	[0-71.2]	15	12538	13 [12-14.1]	99.4 [99.3-99.5]	<0.001	
Sample types								0.604
Dried blood spots	17.1 [14.8-19.5]	NA	1	978	NA	NA	1	
Serum	19 [11.6-27.6]	[0-66.4]	21	14600	11.8 [11-12.7]	99.3 [99.2-99.4]	<0.001	
Recent infection								
Study Design								0.006
Cross sectional	4.3 [2.1-7.2]	[0-24.1]	25	28353	8.2 [7.5-8.9]	98.5 [98.2-98.7]	<0.001	
Outbreak	17.5 [8.1-29.4]	[0-67.4]	5	914	3.9 [2.8-5.4]	93.4 [87.5-96.5]	<0.001	
Sampling								0.622

	Prevalence. % (95%CI)	95% Prediction interval	N Studies	N Participants	H (95%CI)	I^2 (95%CI)	P heterogeneity	P difference subtypes
Non probabilistic	5.7 [3.1-9]	[0-28.9]	27	27752	8.3 [7.7-9]	98.6 [98.3-98.8]	<0.001	
Probabilistic	9.9 [0.2-30.4]	[0-100]	3	1515	9.1 [7-11.8]	98.8 [97.9-99.3]	<0.001	
Setting								0.328
Community-based	8.3 [4.7-12.8]	[0-31.9]	17	11404	6.5 [5.8-7.3]	97.6 [97-98.1]	<0.001	
Hospital-based	4.7 [0.4-12.7]	[0-48.8]	10	17321	11.9 [10.7-13.2]	99.3 [99.1-99.4]	<0.001	
Hospitalization								0.338
Ambulatory	3.5 [0-17.1]	[0-76.8]	6	2945	12.6 [11-14.4]	99.4 [99.2-99.5]	<0.001	
Hospitalized	10.9 [0.4-30.1]	[0-99.3]	4	460	3.2 [2.1-4.8]	90 [77.4-95.6]	<0.001	
Timing of samples collection								0.029
Prospetively	6.3 [3.6-9.6]	[0-29.9]	29	29066	8.4 [7.8-9.1]	98.6 [98.4-98.8]	<0.001	
Retrospectively	2.5 [0.7-5.2]	NA	1	201	NA	NA	1	
Countries								<0.001
Central African Republic	2 [0.2-5.1]	[0-26.9]	4	5965	5.7 [4.3-7.6]	96.9 [94.5-98.3]	<0.001	
Côte d'Ivoire	5.8 [1.4-12.4]	[0-35.2]	6	1161	3.3 [2.5-4.6]	91.1 [83.4-95.2]	<0.001	
Democratic Republic of the Congo	8.4 [5.8-11.4]	NA	1	383	NA	NA	1	
Ethiopia	33.3 [14.5-55.1]	NA	1	21	NA	NA	1	
Guinea	0 [0-3.6]	NA	1	47	NA	NA	1	
Kenya	0 [0-0.1]	NA	2	1337	1 NA	0 NA	0.831	
Nigeria	42.6 [38.7-46.5]	NA	2	635	1 NA	0 NA	0.699	
Senegal	0.6 [0-3.3]	[0-26.2]	4	14780	4.5 [3.2-6.3]	95.1 [90.5-97.5]	<0.001	
Sierra Leone	2.5 [0.7-5.2]	NA	1	201	NA	NA	1	
Sudan	38.1 [24.5-52.6]	[0-91.9]	4	201	1.8 [1.1-3.1]	70.2 [14.4-89.6]	0.018	
Tanzania	0.6 [0.1-1.5]	NA	1	500	NA	NA	1	
Uganda	2.4 [0-16.8]	NA	2	406	2.3 [1.1-4.8]	81.8 [23-95.7]	0.019	
Zambia	0.2 [0.1-0.3]	NA	1	3630	NA	NA	1	
UNSD Region								<0.001
Central Africa	3.1 [0.7-7]	[0-24.8]	5	6348	6.2 [4.9-7.9]	97.4 [95.8-98.4]	<0.001	
Eastern Africa	0.2 [0-1.1]	[0-4.8]	7	5894	2.7 [2-3.8]	86.6 [74.6-93]	<0.001	
Northern Africa	38.1 [24.5-52.6]	[0-91.9]	4	201	1.8 [1.1-3.1]	70.2 [14.4-89.6]	0.018	
West Africa	5.8 [0.8-14.1]	[0-54.4]	14	16824	10.4 [9.4-11.4]	99.1 [98.9-99.2]	<0.001	
Country income level								0.077
Low-income economies	9.3 [4.6-15.2]	[0-38.3]	14	7224	6.2 [5.4-7.1]	97.4 [96.6-98]	<0.001	
Lower-middle income economies	4 [1.2-8.2]	[0-30.1]	16	22043	9.7 [8.9-10.7]	98.9 [98.7-99.1]	<0.001	
YFV vaccine								0.043

	Prevalence. % (95%CI)	95% Prediction interval	N Studies	N Participants	H (95%CI)	I^2 (95%CI)	P heterogeneity	P difference subtypes
No	1.9 [0.2-4.7]	[0-13.7]	7	5660	3.7 [2.8-4.8]	92.6 [87.3-95.7]	<0.001	
Yes/No	21.2 [3.2-48.7]	[0-100]	5	4345	14.2 [12.4-16.3]	99.5 [99.3-99.6]	<0.001	
Study population: Humans								0.001
Apparently healthy individuals	3.5 [0.6-8.1]	[0-24.8]	7	6916	6.3 [5.2-7.6]	97.5 [96.3-98.3]	<0.001	
Febrile patients	7.9 [0.4-22.6]	[0-76.7]	6	16278	15.4 [13.7-17.3]	99.6 [99.5-99.7]	<0.001	
General population	0.6 [0.1-1.5]	NA	1	500	NA	NA	1	
Mixed human categories	15.4 [0-45.5]	[0-100]	3	240	3.3 [2.1-5.4]	91 [76.7-96.6]	<0.001	
YFV positive case contact	8.4 [0-33.3]	NA	2	348	6 [3.8-9.6]	97.3 [93-98.9]	<0.001	
YFV suspected cases	6 [2.2-11.1]	[0-30.4]	11	4985	4.7 [3.9-5.6]	95.5 [93.5-96.8]	<0.001	
Detection assay								<0.001
Enzyme immunoassay	0 [0-0.1]	NA	1	1762	NA	NA	1	
Indirect ELISA	7.2 [3.5-12]	[0-38.8]	24	22373	8.8 [8.1-9.5]	98.7 [98.5-98.9]	<0.001	
Plaque reduction neutralization test (PRNT)	3.7 [2.2-5.4]	NA	1	546	NA	NA	1	